<u>18S V4 Metabarcoding library preparation using two-step PCR for Illumina</u>

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BACKGROUND:

The 18S V4 metabarcoding protocol detailed here is designed to prepare 18S V4 fragments for sequencing on Illumina platforms using a two-step amplification process. The first step (which we call the amplification PCR) will amplify your region of interest and add an iTru tail. The tail acts as the priming site for the second step (which we call the indexing PCR) that will add iTru indices with the Illumina-required sequencing adapter to your product. The resulting libraries are dual-indexed and can be sequenced on any Illumina sequencer.

This protocol is designed such that it can be modified for any organism and/or gene by using custom primers in place of the 18S primers for the amplification PCR, as long as the primers also contain the appropriate iTru tail and the amplicon is an appropriate size for Illumina sequencing. To modify, replace the 18S forward and 18S reverse portions of the primer (in the case below, TAReuk454FWD1 and V4_18SN.Rev, respectively) with your specific forward and reverse primers. <u>Please note – if your primers have inosines, you must use regular Taq (such as GoTaq) instead of a high fidelity, proof-reading DNA polymerase.</u> If you are interested in a different locus and/or your primers do not contain inosine, you may wish to use a high fidelity enzyme such as Kapa HiFi instead. In this case, adjust thermocycler and reaction conditions accordingly.

For running these libraries on an Illumina sequencer, we recommend spiking 10-20% PhiX for MiSeq runs or 20-40% for NextSeq runs to the low complexity of amplicons <u>if your primers did not include heterogeneity spacers.</u> If your locus-specific primers include heterogeneity spacers and/or you're pooling libraries from multiple loci, then the PhiX spike-in can be decreased to 5-10% (or whatever the chosen sequencing core recommends for medium diversity libraries).

For Smithsonian researchers: LAB stocks four plates of unique iTru i5 indices and four plates of unique iTru i7 indices; this is enough to uniquely dual index 384 samples (4 plates), such that no i5 or i7 is repeated within the set. These iTru indexing primers are <u>available for purchase</u> to our LAB users; contact Katie Murphy or a member of LAB staff for details.

Please feel free to reach out to the LAB genomics specialist, Katie Murphy for assistance.

18S PRIMERS (AMPLIFICATION PCR):

- Primer design (fields separated by spaces below)
    - 1. iTru tail 1 → this must be on any custom primers for this protocol; based on TruSeq chemistry. Do NOT edit this portion of the primer sequences.
        - Forward i5: ACACTCTTTCCCTACACGACGCTCTTCCGATCT
        - Reverse i7: GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT
    - 2. Heterogeneity Spacer (0-8bp sequence) – optional but highly recommended to use multiple spacers *of varying length*. This allows for decreased PhiX spike-in and improves sequencing run performance.
        - We suggest using heterogeneity spacers designed by Glenn et al. 2019 (https://baddna.uga.edu/).
    - 3. 18S V4 primer sequence (or other taxon/locus specific sequence)→ this can be altered to fit your needs.
        - Forward (TAReuk454FWD1; Stoeck et al 2010): CCAGCASCYGCGGTAATTC*C
        - Reverse (V4_18SN.Rev; Piredda et al 2017): ACTTTCGTTCTTGATYRATG*A

- These primer examples show the full iTru-tailed 18S V4 primer design. They use the locus primer sequences TAReuk454FWD1 (Stoeck et al 2010) and V4_18SNext.Rev (Piredda et al 2017) and include heterogeneity spacers based on Glenn et al (2019). These are the primers that LAB typically uses; contact Katie Murphy for more information.
    - **Forward:** i5 (F) iTru_18S_V4-F – a mix of the 5 primers below

      ACACTCTTTCCCTACACGACGCTCTTCCGATCT CCAGCASCYGCGGTAATTC*C
      ACACTCTTTCCCTACACGACGCTCTTCCGATCT GGTAC CCAGCASCYGCGGTAATTC*C
      ACACTCTTTCCCTACACGACGCTCTTCCGATCT CAACAC CCAGCASCYGCGGTAATTC*C
      ACACTCTTTCCCTACACGACGCTCTTCCGATCT ATCGGTT CCAGCASCYGCGGTAATTC*C
      ACACTCTTTCCCTACACGACGCTCTTCCGATCT TCGGTCAA CCAGCASCYGCGGTAATTC*C

    - **Reverse:** i7 (R) iTru_18S_V4-R – a mix of the 5 primers below

      GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT ACTTTCGTTCTTGATYRATG*A
      GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT ATCTG ACTTTCGTTCTTGATYRATG*A
      GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT GAGACT ACTTTCGTTCTTGATYRATG*A
      GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT CGATTCC ACTTTCGTTCTTGATYRATG*A
      GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT TCTCAATC ACTTTCGTTCTTGATYRATG*A

    - **Product size:** approx. ~483-499 bp (~416 bp V4 + 67 to 83 bp tails)
        - The 18S V4 fragment is ~375 bp after bioinformatically excluding the primer sites.

- If you are adapting this protocol to use with alternative locus primers and will be using a proofreading enzyme (e.g. KAPA HiFi), you'll need to add a phosphorothioate bond between the last two bases of each primer. The primers above already include this bond (indicated by *).

INDEX PRIMERS (INDEXING PCR):

- For LAB users: LAB has iTru indices available in 8 sets of pre-aliquoted plates (10 µl/well at 10 µM). There are 4 plates of i5 and 4 plates of i7. This is enough for 384 unique dual indices, or 147,456 combinatorial pairs. *We highly recommend using unique dual indices over combinatorial as this will eliminate issues due to index hopping.*

- Each sample needs one i7 and one i5 index. While each index can be used for multiple samples if you use combinatorial indexing, the combination of i7 and i5 must be unique to each sample you plan to pool. However, it's even better to use unique dual indices (UDIs) when possible, such as the iTru set described below, as this mitigates any complications due to index hopping.

- This PCR will increase your fragment size by 69 bp (e.g. the final library size using our 18S V4 primers is ~553-568 bp with spacers).

- Index primer design (fields separated by spaces below)
  - 1. Illumina adapter sequence
  - 2. i7 or i5 index sequence (shown in **bold**)
  - 3. iTru tail 2

- Example Index Primer sequences:

Type	i5 or i7	Name	Sequence
iTru	i7	iTru7_101_01	CAAGCAGAAGACGGCATACGAGAT **GGTAACGT** GTGACTGGAGTTCA*G
iTru	i5	iTru5_01_A	AATGATACGGCGACCACCGAGATCTACAC **ACCGACAA** ACACTCTTTCCCTA*C

* indicates a phosphorothioate bond.

- Please refer to the "LAB Indexing Primers 2024" excel file or Glenn et al 2019 for a full list of available iTru indexing sequences.

SUPPLIES:

- 10 µM iTru-tailed locus-specific primers (forward & reverse)
- 10 µM iTru index primers (i5 & i7)
- 2X HotStart GoTaq master mix
  - *Please remember to adjust your PCR protocols accordingly if you choose to use a different polymerase and/or use an alternative locus.*
- BSA (20 mg/ml)
- Sterile 0.2 mL PCR tubes or plates
- Sterile silicone sealing mats or films (if using PCR plates)
- Sterile 1.5 mL tubes
- KAPA Pure beads (and/or ExoSap Express)
- Ethanol
- Sterile 10mM Tris, pH 8.0
  - Alternatively, you can use TLE (10 mM Tris-HCl, 0.1 mM EDTA). *Make sure to NOT use standard 1X TE, as the higher level of EDTA can interfere with sequencing.*
- Sterile Nuclease-free water
- HS Qubit or Quant-iT assay & tubes
- Filter pipette tips

PROTOCOL:

Make sure to thoroughly clean your bench with 10% bleach (or lab disinfectant) followed by 70% EtOH prior to beginning work each day. Also, make sure to use filter tips.

Step 1 (Optional) – Quantification & dilution of DNA

Quantification and dilution is not mandatory, especially if you are amplifying a large number of samples. However, not quantifying may affect amplification success and product concentration.

1.1 Quantify your DNA sample(s) using QuBit or Quant-iT.
1.2 Normalize your DNA to desired concentration.

Step 2 – 18S V4 PCR

This protocol describes the use of triplicate reactions. Please note that your project may require more replicates than this. We strongly suggest you review recent literature when planning your project.

2.1 Retrieve reagents from the freezer and allow them to thaw at room temperature. Vortex and spin down.
2.2 Perform the initial PCR ***in triplicate*** using 10 µL reactions. *Make sure to include negative PCR controls (PCR blanks).* Follow the recipe below to make your master mix, scaling up accordingly. Volumes are given in µL. Vortex the final master mix thoroughly, then spin briefly.

Component	**Per rxn**	**Master mix:**	**Final concentration**
water	3.3 µL		
2X HotStart GoTaq mix	5.0 µL		1X
10 µM iTru_18S_V4-F primer	0.3 µL		0.3 µM
10 µM iTru_18S_V4-R primer	0.3 µL		0.3 µM
BSA (20 mg/ml)	0.1 µL		0.2 mg/mL
Template DNA	1.0 µL	-----	varies
total	**10 µL**		

2.3 Aliquot 9.0 µL of master mix into each PCR tube or well, then add 1.0 µL of template DNA. Close tubes or cover plate with a sterile silicone mat, then spin to collect liquid to the bottom of wells.
2.4 Place tubes/plate into a thermocycler and run the following program with a heated lid:

95°C – 7 min
35 cycles of:
 95°C – 30 sec
 50°C – 30 sec
 72°C – 30 sec
72°C – 5 min
12°C – hold

2.5 After the PCR finishes, go directly to Step 3, or samples may be stored at 4°C for up to 3 days (or at -20°C for longer periods).

Step 3 – Gel verification & Pooling

3.1 Prepare a 1.5% agarose gel using 1X SB for each plate of PCRs. Each gel should have 8 tiers if you're running PCR plates.
3.2 Mix 2 µl PCR product from Step 2.6 with 2 µl 2X loading dye/10X GelRed. Run gels for 5-7 minutes at 125V.
 a. If you need to make more 2X loading dye/10X GelRed: add 1 µl GelRed stock into 1 mL of 2X loading dye. Mix thoroughly.
3.3 Visualize and/or image the gels to determine the presence/absence of PCR products. Record wells that show incorrect bands.
 a. Product size should be ~483-499 bp for our 18S-V4 primers with heterogeneity spacers.
3.4 Pool PCR replicates for each sample, omitting any that had incorrect bands present.
3.5 Go directly to Step 4, or samples may be stored at 4°C for up to 3 days (or at -20°C for longer periods).

Step 4 – PCR Cleanup – choose 4A or 4B

Clean up your PCR products using either KAPA Pure beads (Step 4A) or ExoSap Express (Step 4B). If you have < 24 µl per sample, please scale the amount of beads or ExoSap accordingly. If you are using a different locus, you may need to adjust the bead ratio based on your amplicon size; refer to bead manufacturer manual for suggestions.

Step 4A – Cleanup with KAPA Pure Beads

4A.1 Remove KAPA Pure beads from the fridge and allow to come to room temperature. While you wait, make fresh 80% Ethanol (450 µl per sample).
4A.2 Vortex the beads until fully resuspended (~30 seconds).
4A.3 To 24 µL of 18S V4 PCR product, add 28.8 µL of KAPA Pure beads (1.2X bead-to-sample ratio).
4A.4 Mix thoroughly by pipetting up/down at least 10 times.
4A.5 Incubate at room temperature for 5 minutes to bind the DNA to the beads.
4A.6 Place the plate/tubes on a magnet for ~3 minutes, or until the supernatant is clear and beads are pelleted.
4A.7 Carefully remove and discard the supernatant, making sure to not disturb the pellet.
4A.8 With the plate/tubes still on the magnet, wash the beads by adding 200µL of 80% ethanol. Do not mix.
4A.9 Incubate at room temp for 30 seconds.
4A.10 Remove and discard the supernatant.
4A.11 Repeat the ethanol wash (steps 4A.8-10).
4A.12 Briefly remove the plate or tubes from the magnet and spin down for a few seconds, then return to the magnetic rack.
4A.13 Carefully remove any residual ethanol using a P20 pipette.
4A.14 Allow beads to airdry for 3 minutes.
4A.15 Remove the plate from the magnet.
4A.16 Add 22 µL of 10 mM Tris-HCl (pH 8.0) (or TLE: 10 mM Tris-HCl, 0.1 mM EDTA) to each well and use a pipette to fully resuspend the beads.
4A.17 Incubate the plate at room temperature for 5 minutes.
4A.18 Place the plate back on the magnet for ~3 minutes, or until the supernatant clears.
4A.19 Carefully transfer 20 µL of the clear supernatant to a new plate/tube, making sure to not disturb the bead pellet.
4A.20 After the cleanup, go directly to Step 5, or samples may be stored at 4°C for up to 3 days (or at -20°C for longer periods).

Step 4B – Cleanup with ExoSap Express

Traditional ExoSap may somewhat degrade your product when you used high fidelity polymerase in your PCR. If you used HiFi or other high fidelity polymerase, use ExoSap express or a bead clean instead of traditional ExoSap. Please note, LAB no longer stocks traditional ExoSap (we only have Express).

4B.1 Dilute your ExoSap Express stock five-fold with water (eg. 1 µL ExoSap + 4 µL water). Make sure to make enough for all of your reactions; you'll need 6 µL per 24 µL PCR product.
4B.2 Add 6 µL diluted ExoSap to 24 µL pooled PCR product. Your final volume in each well should be 30 µL.
4B.3 Pipette to mix, seal the plate, and spin briefly.
4B.4 Place into the thermocycler and run the following program with a heated lid:
    37°C – 4 min
    80°C – 1 min
    12°C – hold
4B.5 After the cleanup, go directly to Step 5, or samples may be stored at 4°C for up to 3 days (or at -20°C for longer periods).

Step 5 (Optional) – Quantification & Dilution

Step 5 is not mandatory, especially if you are amplifying a large number of samples, and we often omit this step at LAB. However, not quantifying may affect amplification success and product concentration. To compare results between samples, it's always ideal to normalize concentrations.

5.1 Use Qubit or Quant-iT to determine the concentration of each sample.

5.2 If needed, dilute your samples to 20 ng/uL or lower in 10mM Tris-HCl (pH 8.0).

   a. You can also normalize all samples to the same concentration, though we find it's often not necessary for this protocol.

5.3 After quantification & dilution, go directly to Step 6, or samples may be stored at 4°C for up to 3 days (or at -20°C for longer periods).

Step 6 – Indexing PCR

6.1 Determine which indices will be used for which samples; make sure to record these exactly. Each sample needs a unique i7 and i5.

6.2 Retrieve the required PCR reagents from the freezer and allow them to thaw at room temperature. Vortex and spin briefly.

6.3 Set up the indexing PCR using 25 µL reactions. Make sure you remember to include an indexing blank (in addition to your earlier PCR blank). Follow the recipe below to make your master mix, scaling up accordingly. Mix thoroughly then spin briefly.

   a. *Remember, your indices are sample-specific; do **NOT** include them in your master mix.*

Component	Per reaction	Master mix	Final Concentration
water	8.75 µL		
2X HotStart GoTaq mix	12.5 µL		1X
10 µM i7 index	0.75 µL	-----	0.3 µM
10 µM i5 index	0.75 µL	-----	0.3 µM
BSA (20 mg/ml)	0.25 µL		0.2 mg/mL
template (18S V4 product)	2.0 µL	-----	varies
total	25 µL		

6.4 If using a master mix (includes everything ***except indices and template***), pipette 21.5 µL of master mix into each well of a new plate.

6.5 Add 0.75 µL of the required i5 index primer to each well, then add 0.75 µL of the required i7 index primer to each well.

   a. Note – if your i5 and i7 were pre-combined, simply add 1.5 µL total.

6.6 Lastly, add 2 µL of your template DNA (18S V4 PCR product). Your total reaction volume in each well should be 25 µL.

6.7 Using a P20 or P200 pipette, set to ~15 µL, mix up/down 5 times. *Avoid making bubbles!*

6.8 Seal the plate with a silicone mat or plate seal and spin briefly to collect liquid at the bottom of the wells.

6.9 Place the plate or tubes into a thermocycler and run the following indexing program with 5-10 cycles, using a heated lid.

95°C – 5 min
5-10 cycles of: *(8 cycles is a good starting point)*
    95°C – 30 sec
    55°C – 30 sec
    72°C – 30 sec
72°C – 5 min
12°C – hold

6.10 After the PCR finishes, go directly to Step 7, or samples may be stored at 4°C for up to 3 days (or at -20°C for longer periods).

Step 7 – Gel verification

7.1 Prepare a 1.5% agarose gel.

7.2 Mix 2 µl PCR product from Step 6 with 2 µl 2X loading dye/10X GelRed. Run gels for 6-8 minutes at 125V.

7.3 Visualize and/or image the gels to determine the presence/absence of indexed PCR products. Record wells that show incorrect bands.

a. Product size should be ~553-568 bp for correctly indexed 18S V4 libraries with heterogeneity spacers. A second band at ~490 bp indicates that not all of your 18S PCR product was indexed.

7.4 Go directly to Step 8, or store samples at 4°C for up to 3 days (or -20°C for longer periods).

Step 8 – PCR Cleanup – choose 8A or 8B

Clean up your PCR products using either KAPA Pure beads (Step 8A) or ExoSap (Step 8B). If you have less than 23 uL per sample, please scale the amount of beads or ExoSap accordingly. Note – if you use ExoSap (Step 8B) at this stage,then you ***must*** *clean-up your final pool with Kapa Pure beads (Step 10) before submitting it for sequencing; the sequencing run will fail if there has been no bead cleanup step. If you follow Step 8A, then you may not need to clean the final pool with beads.*

Step 8A – Cleanup with Kapa Pure Beads

If you are using this protocol for amplicons other than 18S V4, you may need to adjust the bead-to-sample ratio.

8A.1 Remove KAPA Pure beads from the fridge and allow to come to room temperature. While you wait, make fresh 80% Ethanol (~450 µl per sample).

8A.2 Vortex the beads until fully resuspended (~30 seconds).

8A.3 In your indexing PCR plate, add 18 µl of KAPA Pure beads to the ~23 µL of PCR product. This represents a ~0.8X bead-to-sample ratio.

a. This ratio is dependent upon the size of the product; refer to KAPA Pure manual.

8A.4 Mix thoroughly by pipetting at least 10 times, until the mixture is homogenous.

8A.5 Incubate at room temperature for 5 minutes to bind the DNA to the beads.

8A.6 Place the plate/tubes on a magnet for 3 minutes, or until the supernatant is clear and beads are pelleted.

8A.7 Carefully remove and discard the supernatant, making sure to not disturb the pellet.

8A.8 With the plate/tubes still on the magnet, wash the beads by adding 200 µL of 80% ethanol. Do not mix.

8A.9 Incubate at room temp for 30 seconds.

8A.10 Remove and discard the supernatant.

8A.11 Repeat the ethanol wash (steps 8A.8-10).

8A.12 Briefly remove the plate or tubes from the magnet and spin down for a few seconds, then return to the magnetic rack.

8A.13 Carefully remove any residual ethanol using a P20 pipette.

8A.14 Let the plate dry on the magnetic for 3 minutes (*not longer!)*.

8A.15 Remove the plate from the magnet.

8A.16 Add 22 µL of 10 mM Tris-HCl (or TLE: 10 mM Tris-HCl, 0.1 mM EDTA) to each well and use a pipette to fully resuspend the beads.

8A.17 Incubate the plate at room temperature for 5 minutes.

8A.18 Place the plate back on the magnet for ~3 minutes, or until the supernatant clears.

8A.19 Carefully transfer 20 µL of the clear supernatant to a new plate/tube.

8A.20 After the cleanup, go directly to Step 9, or samples may be stored at 4°C overnight (or at -20°C for longer periods).

Step 8B – Cleanup with ExoSap Express

Traditional ExoSap may somewhat degrade your product when you used high fidelity polymerase in your PCR. If you used HiFi or other high fidelity polymerase, use ExoSap express or a bead clean instead of traditional ExoSap. Please note, LAB no longer stocks traditional ExoSap (we only have Express).

8B.1 Dilute your ExoSap stock five-fold with water (eg. 1 µL ExoSap + 4 µL water). Make sure to make enough for all of your reactions. You'll need 5.75 µL per 23 µL PCR product.

8B.2 Add 5.75 µL of diluted ExoSap to each PCR product. Your final volume in each well should now be ~ 28.75 µL.

8B.3 Pipette to mix, seal the plate, and spin briefly.

8B.4 Place into the thermocycler and run the following program with a heated lid:

37C – 4 min
80C – 1 min
12C – hold

8B.5 After the cleanup, go directly to Step 9, or samples may be stored at 4°C for up to 3 days (or at -20°C for longer periods).

Step 9 – Library Quantification & Pooling

9.1 Quantify your libraries using the Qubit or Quant-iT HS dsDNA assay kit, following manufacturer's instructions.

9.2 Pool each library to be sequenced in equal ng or equimolar amounts in a new 1.5 mL tube:
  a. If you are using a single gene and all samples will be approximately the same size (such as with 18S V4), you can pool based on nanograms.
  b. If you are using multiple amplicons that vary in size, you should instead pool in equimolar amounts (nM).

Step 10 – Final Pool Cleanup

This protocol is written for cleaning up a 50 µl aliquot of your pool, but you can adjust this to whatever volume you need. Just make sure to scale the beads proportionally.

*This step is **ONLY** required if you followed Step 8B (ExoSap Cleanup). **Omit** this step if you followed Step 8A (Bead Cleanup), unless you have adapter-dimer concerns. For 18S V4 libraries, we recommend a 0.8X bead-to-sample ratio.*

10.1 Remove KAPA Pure beads from the fridge and allow to come to room temperature. While you wait, make 1mL of fresh 80% Ethanol.

10.2 Vortex the beads until fully resuspended (~30 seconds).

10.3 To a 50 µl aliquot of your pool in a 1.5 mL tube, add 40 µl of KAPA Pure beads. This represents a 0.8X bead-to-sample ratio.
  a. If you wish to clean up a different volume of your pool, make sure to adjust the bead volume accordingly.
  b. This ratio is dependent upon the size of the product; refer to KAPA Pure manual.

10.4 Mix thoroughly by pipetting or vortexing, until the mixture is homogenous.

10.5 Incubate at room temperature for 5-15 minutes to bind the DNA to the beads.

10.6 Place the tube on a magnetic stand for ~3 minutes, or until the supernatant is clear and beads are pelleted.

10.7 Carefully remove and discard the supernatant, making sure to not disturb the pellet.

10.8 With the tube still on the magnet, wash the beads by adding 500 µL of 80% ethanol. Do not mix.

10.9 Incubate at room temp for 30 seconds.

10.10 Remove and discard the supernatant.

10.11 Repeat the ethanol wash (steps 10.8-10).

10.12 Briefly remove the tube from the magnet and spin down for a few seconds, then return to the magnetic rack.

10.13 Carefully remove all residual ethanol using a P20 pipette.

10.14 Let the plate dry on the magnetic for 3 minutes.

10.15 Remove the tube from the magnet.

10.16 Add 50 µL of 10 mM Tris-HCl (or TLE: 10 mM Tris-HCl, 0.1 mM EDTA) to the tube and pipette to fully resuspend the beads.
  a. Adjust the volume of 10 mM Tris if you wish to concentrate your pool.

10.17 While off the magnet, incubate the tube at room temperature for 5 minutes.

10.18 Place the tube back on the magnet for ~3 minutes, or until the supernatant clears.

10.19 Carefully transfer 48 µL of the clear supernatant to a new tube; make sure not to disturb the bead pellet.

Step 11 – Final Pool QC

11.1 Use HS dsDNA Qubit or Quant-iT to measure the concentration (ng/µL) of your library pool.

11.2 Run a 1:10 dilution of your pool on an Agilent HS D1000 ScreenTape to verify the product is the correct size and no adapter-dimer is present.
  a. Alternatively, run 2 uL of your undiluted pool on a 1.5% agarose gel with an appropriate DNA ladder. Suggested settings for mini-gels are 125V for approximately 30 minutes; this will need to

be adjusted for different gel rigs. Visualize the gel to determine if an adapter-dimer band is present. Please note that this will not be as sensitive as a Tapestation or Bioanalyzer.
b. If adapter-dimer is present, we suggest performing an additional 0.8X bead cleanup. Repeat gel and Qubit quantification after any additional cleanups.

11.3 Determine the molarity (nM) of your pool using the equation given in the "Additional Information" section of this guide.

11.4 Your pool should be stored at -20°C until submitted for sequencing.

You're now ready to submit an aliquot of your pool to an Illumina sequencing facility, along with any required documentation.

If you need to ship your pool to a sequencing facility, we recommend shipping it on blue or dry ice.

If the pool will be sequenced at NMNH-LAB, directions for order submission can be found on LAB's website. Reach out to Katie Murphy with any questions.

ADDITIONAL INFORMATION:

- If you need to convert from ng/ul to nM, please use the following equation.

$$\frac{(\textbf{concentration in ng}/\mu\textbf{L})}{(\textbf{660 g/mol}) \times (\textbf{average library size in bp})} \times \mathbf{10^6} = \textbf{concentration in nM}$$

Average library size should be the total length (including index/adapters), not just your insert length.

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